DEPARTMENT OF COMPUTING
AND CONTROL ENGINEERING

ДЕПАРТМАН ЗА РАЧУНАРСТВО
И АУТОМАТИКУ

ACSE

Odsek za automatiku, geomatiku
i upravljanje sistemima

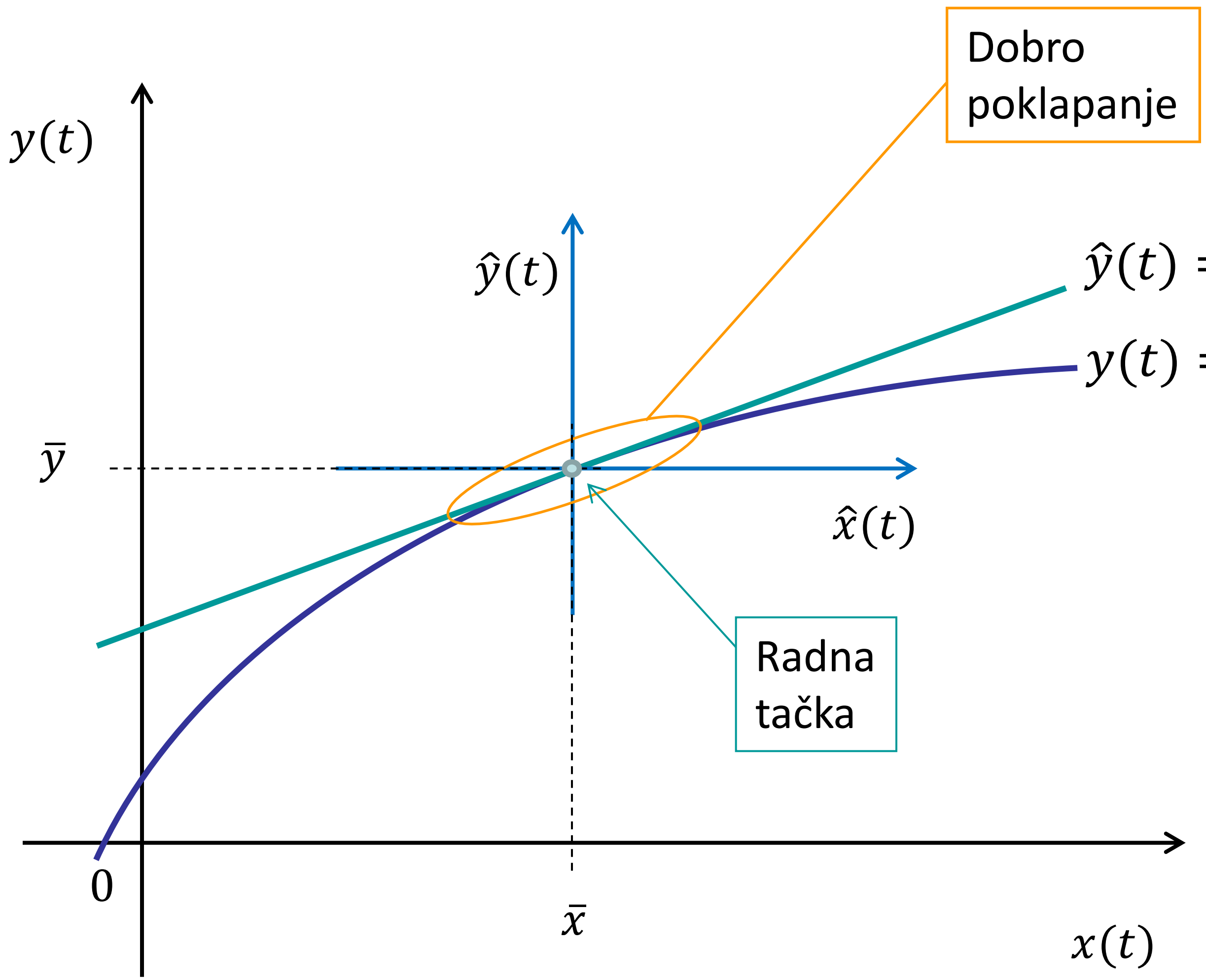
Sub- department for Automatic Control
and System Engineering

Linearizacija matematičkog modela

Modeliranje i simulacija sistema

Upravljanje, modelovanje i simulacija sistema

Princip linearizacije – grafički



$\hat{y}(t) = m \cdot \hat{x}(t)$ ← linearan model

$y(t) = f(x(t))$ ← nelinearan model

$x(t) = \bar{x} + \hat{x}(t)$

$x(t)$ – izmerena (fizička) vrednost
$\bar{x}$ – nominalna vrednost
$\hat{x}(t)$ – inkrementalna vrednost

$y(t) = \bar{y} + \hat{y}(t)$

Radna tačka

- Izbor radne tačke?
  - Vrednosti u ustaljenom stanju = nominalne vrednosti
  - Sistem može menjati ustaljeno stanje
    - Tada se menja radna tačka i za nju važi drugačiji linearan model

- Radnu tačku u modelu čine vrednosti ulaza, promenljivih stanja i izlaza
  - Te vrednosti se predstavljaju zbirom **nominalne** i **inkrementalne vrednosti**

$$x_k(t) = \bar{x}_k + \hat{x}_k(t), \quad k = 1,2,\dots,n$$

$$u_k(t) = \bar{u}_k + \hat{u}_k(t), \quad k = 1,2,\dots,m$$

$$y_k(t) = \bar{y}_k + \hat{y}_k(t), \quad k = 1,2,\dots,r$$

Primer linearizacije

- Nelinearan model:

$$y(t) = a \cdot u(t) + b$$

- U radnoj tački je:

$$y(t) = \bar{y}$$

$$u(t) = \bar{u},$$

tj. važi: $\bar{y} = a \cdot \bar{u} + b$

- Uvode se smene:

$$y(t) = \bar{y} + \hat{y}(t)$$

$$u(t) = \bar{u} + \hat{u}(t)$$

- Nakon zamene u model:

$$\bar{y} + \hat{y}(t) = a\big(\bar{u} + \hat{u}(t)\big) + b = a\bar{u} + a\hat{u}(t) + b$$

ostaje

$$\hat{y}(t) = a \cdot \hat{u}(t)$$

Linearan model

Principip linearizacije – analitički

- Razvoj u Tejlorov red funkcije jedne promenljive u okolini radne tačke (nominalne vrednosti)

$$y(t) = f(x(t))$$

$$y(t) = f(\bar{x}) + \left.\frac{df}{dx}\right|_{\bar{\boldsymbol{x}}} \frac{x(t)-\bar{x}}{1!} + \left.\frac{d^2 f}{dx^2}\right|_{\bar{\boldsymbol{x}}} \frac{(x(t)-\bar{x})^2}{2!} + \left.\frac{d^3 f}{dx^3}\right|_{\bar{\boldsymbol{x}}} \frac{(x(t)-\bar{x})^3}{3!} + \cdots$$

$$y(t) \approx f(\bar{x}) + \left.\frac{df}{dx}\right|_{\bar{\boldsymbol{x}}} \frac{(x(t)-\bar{x})}{1!}$$

- Smene uvode inkrementalne promenljive

$y(t) = \bar{y} + \hat{y}(t)$, tj. $\quad \hat{y}(t) = y(t) - \bar{y},\ \bar{y} = f(\bar{x})$

$x(t) = \bar{x} + \hat{x}(t)$

- Dobija se linearan model

$$\hat{y}(t) = a\hat{x}(t)$$

gde je a konstanta

$$a = \left.\frac{df}{dx}\right|_{\bar{\boldsymbol{x}}}$$

Princip linearizacije – analitički (2)

- Generalizacija razvoja u Tejlorov red za funkciju dve promenljive $y(t) = f(\boldsymbol{x}) = f(x_1, x_2)$

$$y(t) = f(\bar{\boldsymbol{x}}) + \left.\frac{\partial f}{\partial x_1}\right|_{\bar{\boldsymbol{x}}} (x_1 - \bar{x}_1) + \left.\frac{\partial f}{\partial x_2}\right|_{\bar{\boldsymbol{x}}} (x_2 - \bar{x}_2)$$

$$+ \frac{1}{2}\left(\left.\frac{\partial^2 f}{\partial x_1^2}\right|_{\bar{\boldsymbol{x}}} (x_1 - \bar{x}_1)^2 + 2 \left.\frac{\partial^2 f}{\partial x_1 \partial x_2}\right|_{\bar{\boldsymbol{x}}} (x_1 - \bar{x}_1)(x_2 - \bar{x}_2) + \left.\frac{\partial^2 f}{\partial x_2^2}\right|_{\bar{\boldsymbol{x}}} (x_2 - \bar{x}_2)^2 \right) + \cdots$$

$$y(t) \approx f(\bar{\boldsymbol{x}}) + \left.\frac{\partial f}{\partial x_1}\right|_{\bar{\boldsymbol{x}}} (x_1 - \bar{x}_1) + \left.\frac{\partial f}{\partial x_2}\right|_{\bar{\boldsymbol{x}}} (x_2 - \bar{x}_2)$$

- Smene uvode inkrementalne promenljive

$y(t) = \bar{y} + \hat{y}(t)$, tj. $\qquad \hat{y}(t) = y(t) - f(\bar{x}_1, \bar{x}_2)$

$x_1(t) = \bar{x}_1 + \hat{x}_1(t), \qquad x_2(t) = \bar{x}_2 + \hat{x}_2(t)$

- Dobija se linearan model

$$\hat{y}(t) = a_1 \hat{x}_1(t) + a_2 \hat{x}_2(t)$$

$$a_1 = \left.\frac{df}{dx_1}\right|_{\bar{\boldsymbol{x}}}, \ a_2 = \left.\frac{df}{dx_2}\right|_{\bar{\boldsymbol{x}}}$$

Princip linearizacije – analitički (3)

- Generalizacija razvoja u Tejlorov red za funkciju više promenljivih $y(t) = f(\boldsymbol{x}) = f(x_1, x_2, \dots x_n)$

$$y(t) \approx f(\overline{\boldsymbol{x}}) + \left.\frac{\partial f}{\partial x_1}\right|_{\overline{\boldsymbol{x}}} (x_1 - \bar{x}_1) + \left.\frac{\partial f}{\partial x_2}\right|_{\overline{\boldsymbol{x}}} (x_2 - \bar{x}_2) + \cdots + \left.\frac{\partial f}{\partial x_n}\right|_{\overline{\boldsymbol{x}}} (x_n - \bar{x}_n)$$

- Smene uvode inkrementalne promenljive

$y(t) = \bar{y} + \hat{y}(t)$, tj. $\qquad \hat{y}(t) = y(t) - f(\overline{\boldsymbol{x}})$

$x_k(t) = \bar{x}_k + \hat{x}_k(t), k = 1,2, \dots n$

- Dobija se linearan model

$$\hat{y}(t) = a_1\hat{x}_1(t) + a_2\hat{x}_2(t) + \cdots + a_n\hat{x}_n(t)$$

  - $a_1, a_2, \dots a_n$ su konstante.

Nelinearan model u prostoru stanja (još jednom)

- Pomena (kretanje) promenljivih stanja

$$\dot{x}_1(t) = f_1(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_m; t), \quad x_1(t_0) = x_{10}$$
$$\dot{x}_2(t) = f_2(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_m; t), \quad x_2(t_0) = x_{20}$$
$$\dots$$
$$\dot{x}_n(t) = f_n(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_m; t), \quad x_n(t_0) = x_{n0}$$

- Izlazi modela

$$y_1(t) = g_1(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_m; t)$$
$$y_2(t) = g_2(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_m; t)$$
$$\dots$$
$$y_r(t) = g_r(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_m; t)$$

- <u>Linearizacija nelinearnog modela u prostoru stanja</u>
  Nelinearan model sadrži nelinearne funkcije $f_1, f_2, \dots f_n$ i $g_1, g_2, \dots g_r$ koje pojedinačno treba linearizovati prethodno opisanim principom.

Linearizacija (nelinearnog) modela u prostoru stanja

- Linerarizacija funkcije f_i u okolini radne tačke $(\bar{\boldsymbol{x}}, \bar{\boldsymbol{u}})$:

$$\dot{x}_i(t) = f_i(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_m; t), \quad i = 1,2, \dots n$$

$$\dot{x}_i(t) \approx f_i\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}} + \frac{\partial f_i}{\partial x_1}\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}} (x_1 - \bar{x}_1) + \cdots + \frac{\partial f_i}{\partial x_n}\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}} (x_n - \bar{x}_n) + \frac{\partial f_i}{\partial u_1}\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}} (u_1 - \bar{u}_1) + \cdots + \frac{\partial f_i}{\partial u_m}\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}} (u_m - \bar{u}_m)$$

- Smene uvode inkrementalne promenljive:

$$x_k(t) = \bar{x}_k + \hat{x}_k(t), \quad k = 1,2, \dots, n$$

$$u_k(t) = \bar{u}_k + \hat{u}_k(t), \quad k = 1,2, \dots, m$$

$\dot{x}_k(t) = \dot{\bar{x}}_k + \dot{\hat{x}}_k(t)$
Ako je nominalna vrednost konstanta $\dot{\bar{x}}_k = 0$, tj. $\dot{x}_k(t) = \dot{\hat{x}}_k(t)$.

Takođe $f_i|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}} = f_i(\bar{\boldsymbol{x}}, \bar{\boldsymbol{u}}) = 0$ jer je to uslov za određivanje radne tačke.

$$\dot{\hat{x}}_i = \dot{x}_i - f_i|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}} = \frac{\partial f_i}{\partial x_1}\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}} \hat{x}_1 + \cdots + \frac{\partial f_i}{\partial x_n}\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}} \hat{x}_n + \frac{\partial f_i}{\partial u_1}\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}} \hat{u}_1 + \cdots + \frac{\partial f_i}{\partial u_m}\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}} \hat{u}_m$$

$$\dot{\hat{x}}_i(t) = a_{i1}\hat{x}_1(t) + \cdots + a_{in}\hat{x}_n(t) + b_{i1}\hat{u}_1(t) + \cdots + b_{im}\hat{u}_m(t)$$

Sistem običnih linearnih diferencijalnih jednačina 1. reda

- Nakon linearizacije svih n nelinearnih funkcija f dobija se:

$$\dot{\hat{x}}_1(t) = a_{11}\hat{x}_1(t) + \cdots + a_{1n}\hat{x}_n(t) + b_{11}\hat{u}_1(t) + \cdots + b_{1m}\hat{u}_m(t) \qquad \hat{x}_1(0) = \hat{x}_{10}$$
$$\dot{\hat{x}}_2(t) = a_{21}\hat{x}_1(t) + \cdots + a_{2n}\hat{x}_n(t) + b_{21}\hat{u}_1(t) + \cdots + b_{2m}\hat{u}_m(t) \qquad \hat{x}_2(0) = \hat{x}_{20}$$
$$\dots \qquad \dots$$
$$\dot{\hat{x}}_n(t) = a_{n1}\hat{x}_1(t) + \cdots + a_{nn}\hat{x}_n(t) + b_{n1}\hat{u}_1(t) + \cdots + b_{nm}\hat{u}_m(t) \qquad \hat{x}_n(0) = \hat{x}_{n0}$$

gde su konstante

$$a_{ij} = \left.\frac{\partial f_i}{\partial x_j}\right|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}}, \qquad b_{ik} = \left.\frac{\partial f_i}{\partial u_k}\right|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}}, \qquad i = 1,2,\dots,n, \qquad j = 1,2,\dots,n, \qquad k = 1,2,\dots,m$$

Linearizacija (nelinearnog) modela u prostoru stanja (2)

- Linerarizacija funkcije g_i u okolini radne tačke $(\bar{\boldsymbol{x}}, \bar{\boldsymbol{u}}, \bar{\boldsymbol{y}})$:

$$y_i(t) = g_i(x_1, x_2, \dots, x_n; u_1, u_2, \dots, u_m; t), \quad i = 1,2, \dots r$$

$$y_i \approx g_i\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}} + \frac{\partial g_i}{\partial x_1}\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}}(x_1 - \bar{x}_1) + \dots + \frac{\partial g_i}{\partial x_n}\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}}(x_n - \bar{x}_n) + \frac{\partial g_i}{\partial u_1}\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}}(u_1 - \bar{u}_1) + \dots + \frac{\partial g_i}{\partial u_m}\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}}(u_m - \bar{u}_m)$$

- Smene uvode inkrementalne promenljive:

$$x_k(t) = \bar{x}_k + \hat{x}_k(t), \quad k = 1,2, \dots, n$$
$$u_k(t) = \bar{u}_k + \hat{u}_k(t), \quad k = 1,2, \dots, m$$
$$y_k(t) = \bar{y}_k + \hat{y}_k(t), \quad k = 1,2, \dots, r$$

$$\hat{y}_i = y_i - g_i\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}} = \frac{\partial g_i}{\partial x_1}\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}}\hat{x}_1 + \dots + \frac{\partial g_i}{\partial x_n}\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}}\hat{x}_n + \frac{\partial g_i}{\partial u_1}\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}}\hat{u}_1 + \dots + \frac{\partial g_i}{\partial u_m}\Big|_{\substack{\bar{\boldsymbol{x}} \\ \bar{\boldsymbol{u}}}}\hat{u}_m$$

$$\hat{y}_i(t) = c_{i1}\hat{x}_1(t) + \dots + c_{in}\hat{x}_n(t) + d_{i1}\hat{u}_1(t) + \dots + d_{im}\hat{u}_m(t)$$

Sistem linearnih algebarskih jednačina

- Nakon linearizacije svih r nelinearnih funkcija g dobija se:

$$\hat{y}_1(t) = c_{11}\hat{x}_1(t) + \cdots + c_{1n}\hat{x}_n(t) + d_{11}\hat{u}_1(t) + \cdots + d_{1m}\hat{u}_m(t)$$
$$\hat{y}_2(t) = c_{21}\hat{x}_1(t) + \cdots + c_{2n}\hat{x}_n(t) + d_{21}\hat{u}_1(t) + \cdots + d_{2m}\hat{u}_m(t)$$
$$\ldots$$
$$\hat{y}_r(t) = c_{r1}\hat{x}_1(t) + \cdots + c_{rn}\hat{x}_n(t) + d_{r1}\hat{u}_1(t) + \cdots + d_{rm}\hat{u}_m(t)$$

gde su konstante

$$c_{ij} = \left.\frac{\partial g_i}{\partial x_j}\right|_{\substack{\bar{\boldsymbol{x}}\\ \bar{\boldsymbol{u}}}}, \qquad d_{ik} = \left.\frac{\partial g_i}{\partial u_k}\right|_{\substack{\bar{\boldsymbol{x}}\\ \bar{\boldsymbol{u}}}}, \qquad i = 1,2,\ldots,r, \qquad j = 1,2,\ldots,n, \qquad k = 1,2,\ldots,m$$

Linearan matematički model u prostoru stanja (u razvijenom obliku)

$$
\begin{aligned}
\dot{\hat{x}}_1(t) &= a_{11}\hat{x}_1(t) + \cdots + a_{1n}\hat{x}_n(t) + b_{11}\hat{u}_1(t) + \cdots + b_{1m}\hat{u}_m(t) \\
\dot{\hat{x}}_2(t) &= a_{21}\hat{x}_1(t) + \cdots + a_{2n}\hat{x}_n(t) + b_{21}\hat{u}_1(t) + \cdots + b_{2m}\hat{u}_m(t) \\
&\dots \\
\dot{\hat{x}}_n(t) &= a_{n1}\hat{x}_1(t) + \cdots + a_{nn}\hat{x}_n(t) + b_{n1}\hat{u}_1(t) + \cdots + b_{nm}\hat{u}_m(t)
\end{aligned}
\qquad
\begin{aligned}
\hat{x}_1(0) &= \hat{x}_{10} \\
\hat{x}_2(0) &= \hat{x}_{20} \\
&\dots \\
\hat{x}_n(0) &= \hat{x}_{n0}
\end{aligned}
$$

$$
\begin{aligned}
\hat{y}_1(t) &= c_{11}\hat{x}_1(t) + \cdots + c_{1n}\hat{x}_n(t) + d_{11}\hat{u}_1(t) + \cdots + d_{1m}\hat{u}_m(t) \\
\hat{y}_2(t) &= c_{21}\hat{x}_1(t) + \cdots + c_{2n}\hat{x}_n(t) + d_{21}\hat{u}_1(t) + \cdots + d_{2m}\hat{u}_m(t) \\
&\dots \\
\hat{y}_r(t) &= c_{r1}\hat{x}_1(t) + \cdots + c_{rn}\hat{x}_n(t) + d_{r1}\hat{u}_1(t) + \cdots + d_{rm}\hat{u}_m(t)
\end{aligned}
$$

Linearan matematički model u prostoru stanja (u vektorskom obliku)

$$\dot{\hat{\boldsymbol{x}}}(t) = \boldsymbol{A} \cdot \hat{\boldsymbol{x}}(t) + \boldsymbol{B} \cdot \hat{\boldsymbol{u}}(t), \qquad \hat{\boldsymbol{x}}(0) = \boldsymbol{x}_0$$

$$\hat{\boldsymbol{y}}(t) = \boldsymbol{C} \cdot \hat{\boldsymbol{x}}(t) + \boldsymbol{D} \cdot \hat{\boldsymbol{u}}(t)$$

$$\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \quad \boldsymbol{B} = \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \dots & \dots & & \dots \\ b_{n1} & b_{n2} & \dots & b_{nm} \end{bmatrix}$$

$$\boldsymbol{C} = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \dots & \dots & & \dots \\ c_{r1} & c_{r2} & \dots & c_{rn} \end{bmatrix} \quad \boldsymbol{D} = \begin{bmatrix} d_{11} & d_{12} & \dots & d_{1m} \\ d_{21} & d_{22} & \dots & d_{2m} \\ \dots & \dots & & \dots \\ d_{r1} & d_{r2} & \dots & d_{rm} \end{bmatrix}$$

$$\hat{\boldsymbol{u}} = \begin{bmatrix} \hat{u}_1 \\ \hat{u}_2 \\ \dots \\ \hat{u}_m \end{bmatrix}, \qquad \hat{\boldsymbol{y}} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \dots \\ \hat{y}_r \end{bmatrix}, \qquad \hat{\boldsymbol{x}} = \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \dots \\ \hat{x}_n \end{bmatrix}.$$

Pisanje “kapica” se izbegava

Određivanje radne tačke

- nominalne vrednosti (promenljivih stanja, ulaza i izlaza) u radnoj tački su nekada unapred poznate
  - mogu se dobiti merenjem
  - deo tih vrednosti se može izračunati na osnovu poznavanja ostalih nominalnih vrednosti

- Nominalne vrednosti ulaza se znaju $\bar{u}_1, \bar{u}_2, \dots, \bar{u}_m$
- Reši se n algebarskih jednačina da se nađu $\bar{x}_1, \bar{x}_2, \dots . \bar{x}_n$

$$0 = f_1(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n; \bar{u}_1, \bar{u}_2, \dots, \bar{u}_m)$$
$$0 = f_2(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n; \bar{u}_1, \bar{u}_2, \dots, \bar{u}_m)$$
$$\dots$$
$$0 = f_n(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n; \bar{u}_1, \bar{u}_2, \dots, \bar{u}_m)$$

- Izračunaju se nominalne vrednosti izlaza

$$\bar{y}_1 = g_1(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n; \bar{u}_1, \bar{u}_2, \dots, \bar{u}_m)$$
$$\bar{y}_2 = g_2(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n; \bar{u}_1, \bar{u}_2, \dots, \bar{u}_m)$$
$$\dots$$
$$\bar{y}_r = g_r(\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n; \bar{u}_1, \bar{u}_2, \dots, \bar{u}_m)$$

Primetiti da su ove jednačine nastale iz $\dot{x}_k(t) = 0, k = 1,2, \dots n$, jer se radna tačka „mirna“, tj. nema promena za nominalne vrednosti.

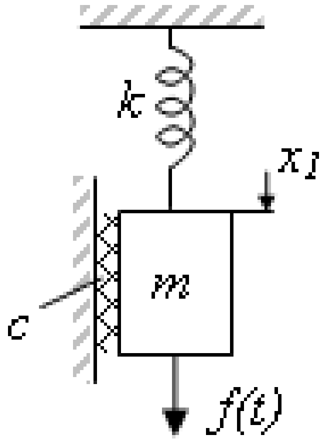
Koraci linearizacije

Koraci postupka linearizacije su:

1. Odrediti radnu tačku – pisanjem i rešavanjem odgovarajućih algebarskih jednačina (prethodni slajd)
2. Prepisati sve linearne članove kao sume nominalne i inkrementalne vrednosti
3. Zameniti sve nelinearne članove sa prva 2 sabirka razvoja u Tejlovor red
4. Skratiti konstantne članove u differencijalnim jednačinama
   (upotrebiti algebarske jednačine koje ordeđuju radnu tačku)
5. Definisati početne vrednosti inkrementalnih promenljivih $\hat{x}(0) = x(0) - \bar{x}$

Primer: Amortizer

Amortizer se kreće pod dejstvom gravitacione sile i pobudne sile $f(t)$. Međutim, ne može se smatrati da je sila u opruzi linearna za sva istezanja, nego je ona funkcija pomeraja x, $f_k = fk(x)$. Ukoliko je ta sila $f_k(x) = kx^3$, $k = const$ i pobudna sila $\bar{f} = const$ odrediti vrednost pomeraja $\bar{x}$. Nadalje, ako se pobudna sila menja kao $f(t) = \bar{f} + \hat{f}(t)$, napisati linearan matematički model koji opisuje promene položaja u odnosu na nominalnu vrednost x_0.



$$m\ddot{x} + c\dot{x} + f_k(x) = mg + f(t)$$

Primer: Amortizer (2)

- Korak 1: u ustaljenom stanju, promene položaja x ne postoje i tada važi

$$k\bar{x}^3 = mg + \bar{f}, \text{ tj. } \bar{x} = \sqrt[3]{\frac{mg+\bar{f}}{k}}$$

- Korak 2: smene: $f(t) = \bar{f} + \hat{f}(t), x(t) = \bar{x} + \hat{x}(t)$

$$m\ddot{\hat{x}} + c\dot{\hat{x}} + k(\bar{x} + \hat{x})^3 = mg + \bar{f} + \hat{f}(t)$$

- Korak 3:

Linearizacija nelinearnog člana $f_k(x) = kx^3$

$$f_k(\bar{x} + \hat{x}(t)) \approx f(\bar{x}) + \left.\frac{\partial f(x)}{\partial x}\right|_{x=\bar{x}} \hat{x}(t) = k\bar{x}^3 + 3k\bar{x}^2\hat{x}(t)$$

Nakon zamene

$$m\ddot{\hat{x}} + c\dot{\hat{x}} + k\bar{x}^3 + 3k\bar{x}^2\hat{x} = mg + \bar{f} + \hat{f}$$

- Korak 4:

$$m\ddot{\hat{x}} + c\dot{\hat{x}} + k_1\hat{x} = \hat{f}(t), \quad k_1 = 3k\bar{x}^2$$

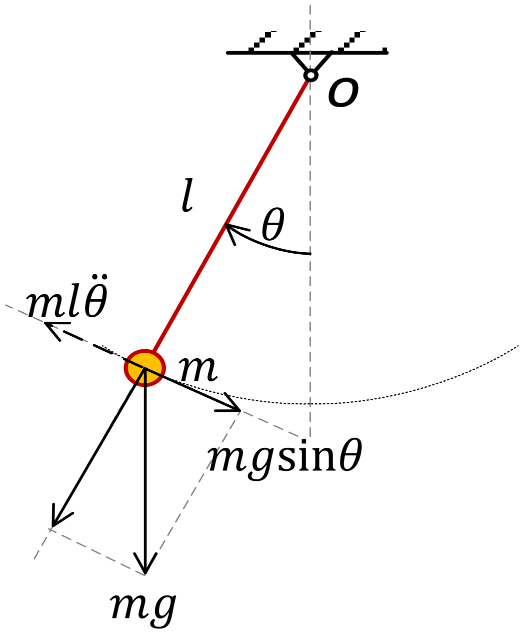
- Korak 5:

$$\hat{x}(0) = x(0) - \bar{x}, \dot{\hat{x}}(0) = \dot{x}(0)$$

Primer: Matematičko klatno

Model: $\ddot{\theta}(t) + \frac{g}{l}\sin(\theta(t)) = 0$

$\theta(0) = a, \dot{\theta}(0) = b$



<u>Linearizacija</u>

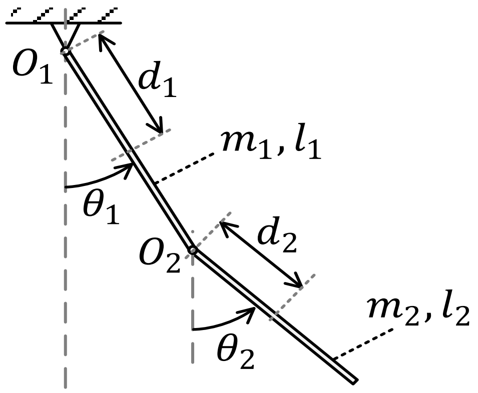
1. Radna tačka je: $\bar{\theta} = 0$
2. Smena: $\theta(t) = \bar{\theta} + \hat{\theta}(t)$
3. Razvoj u Tejlorov red: $\sin(\bar{x} + \hat{x}) = \sin(\bar{x}) + \cos(\bar{x})\,\hat{x}$

   $\ddot{\hat{\theta}}(t) + \frac{g}{l}\cos(\bar{\theta})\hat{\theta}(t) = 0$
4. Skraćivanje: $\ddot{\hat{\theta}}(t) + \frac{g}{l}\hat{\theta}(t) = 0$
5. Početna vrednosti: $\hat{\theta}(0) = \theta(0) - \bar{\theta} = a, \dot{\hat{\theta}}(0) = \dot{\theta}(0) = b$

Do istog rešenja se dolazi kada se pretpostavi da je $\sin(\theta) \approx \theta$

Primer: Dvostepeno fizičko klatno od štapova

Model:



$$\ddot{\theta}_1 + \frac{3}{8}\left[\ddot{\theta}_2 \cos(\theta_1 - \theta_2) + \dot{\theta}_2^2 \sin(\theta_1 - \theta_2)\right] + \frac{9g}{8l}\sin\theta_1 = 0$$

$$\ddot{\theta}_2 + \frac{3}{2}\left[\ddot{\theta}_1 \cos(\theta_1 - \theta_2) - \dot{\theta}_1^2 \sin(\theta_1 - \theta_2)\right] + \frac{3g}{2l}\sin\theta_2 = 0$$

ili

$$\dot{\theta}_1 = \omega_1$$

$$\dot{\omega}_1 = \frac{-4a^2\omega_1^2 CS - a\omega_2^2 S + \frac{4}{3}abC\sin\theta_2 - b\sin\theta_1}{1 - 4a^2C^2}$$

$$\dot{\theta}_2 = \omega_2$$

$$\dot{\omega}_2 = \frac{4a\omega_1^2 S + 4a^2\omega_2^2 CS + 4abC\sin\theta_1 - \frac{4}{3}b\sin\theta_2}{1 - 4a^2C^2}$$

$a = \frac{3}{8}$
$b = \frac{9}{8}\frac{g}{l}$
$C = \cos(\theta_1 - \theta_2)$
$S = \sin(\theta_1 - \theta_2)$

Linearizacija

1. Radna tačka je: $\bar{\theta}_1 = \bar{\theta}_2 = \bar{\omega}_1 = \bar{\omega}_2 = 0$
2. Smene: $\theta_1(t) = \bar{\theta}_1 + \hat{\theta}_1(t)$, $\theta_2(t) = \bar{\theta}_2 + \hat{\theta}_2(t)$, $\omega_1(t) = \bar{\omega}_1 + \hat{\omega}_1(t)$, $\omega_2(t) = \bar{\omega}_2 + \hat{\omega}_2(t)$

…

Primer: Dvostepeno fizičko klatno od štapova (2)

$\dot{\omega}_1 = f_1(\theta_1, \omega_1, \theta_2, \omega_2) = f_1(\boldsymbol{x})$ i $\dot{\omega}_2 = f_2(\theta_1, \omega_1, \theta_2, \omega_2) = f_2(\boldsymbol{x}),\ \boldsymbol{x} = [\theta_1\ \omega_1\ \theta_2\ \omega_2]^T$

Tražimo izvode: $\frac{\partial f_1}{\partial \theta_1}, \frac{\partial f_1}{\partial \omega_1}, \frac{\partial f_1}{\partial \theta_2}, \frac{\partial f_1}{\partial \omega_2}$, kao i $\frac{\partial f_2}{\partial \theta_1}, \frac{\partial f_2}{\partial \omega_1}, \frac{\partial f_2}{\partial \theta_2}, \frac{\partial f_2}{\partial \omega_2}$

Računamo:

$$\dot{\widehat{\omega}}_1 = \frac{\partial f_1(\overline{\boldsymbol{x}})}{\partial \theta_1}\hat{\theta}_1 + \frac{\partial f_1(\overline{\boldsymbol{x}})}{\partial \omega_1}\widehat{\omega}_1 + \frac{\partial f_1(\overline{\boldsymbol{x}})}{\partial \theta_2}\hat{\theta}_2 + \frac{\partial f_1(\overline{\boldsymbol{x}})}{\partial \omega_2}\widehat{\omega}_2, \text{ gde je } \overline{\boldsymbol{x}} = \left[\bar{\theta}_1\ \overline{\omega}_1\ \bar{\theta}_2\ \overline{\omega}_2\right]^T$$

$$\dot{\widehat{\omega}}_2 = \frac{\partial f_2(\overline{\boldsymbol{x}})}{\partial \theta_1}\hat{\theta}_1 + \frac{\partial f_2(\overline{\boldsymbol{x}})}{\partial \omega_1}\widehat{\omega}_1 + \frac{\partial f_2(\overline{\boldsymbol{x}})}{\partial \theta_2}\hat{\theta}_2 + \frac{\partial f_2(\overline{\boldsymbol{x}})}{\partial \omega_2}\widehat{\omega}_2$$

(Ovde ima „dosta posla“!)

Konačno:

$$\dot{\theta}_1 = \omega_1$$

$$\dot{\omega}_1 = -\frac{18}{7}\frac{g}{l}\theta_1 + \frac{9}{7}\frac{g}{l}\theta_2$$

$$\dot{\theta}_2 = \omega_2$$

$$\dot{\omega}_2 = \frac{27}{7}\frac{g}{l}\theta_1 - \frac{24}{7}\frac{g}{l}\theta_2$$

Klatno: Poređenje odziva nelinearnog i linearnog modela

- Slike prikazuju vremenske promene uglova štapova:
  - Nelinearan model plava (θ_1) i crvena kriva (θ_2)
  - Linearan model žuta ($\hat{\theta}_1$) i ljubičasta kriva ($\hat{\theta}_2$)
- za različita početna stanja:
  - Levo $\theta_1(0) = 0.1$, sredina $\theta_1(0) = 0.2$, desno $\theta_1(0) = 1$
  - Ostale promenljive stanja su 0 u $t = 0$.